

AXIOM AI TECHNOLOGIES

Disaster Recovery Plan

Document Control

Field	Value
Document ID	DRP-001
Version	1.0
Status	Portfolio Demonstration
Classification	Internal Use
Document Owner	Information Security / GRC
Technical Owner	Engineering / IT
Business Owner	Executive Management
Approval Authority	Executive Management
Review Frequency	At least annually and following significant incidents, tests, or material changes
Related Documents	BIA; BCP/DR Strategy; Business Continuity Plan; Incident Response Policy

Portfolio Disclaimer: Axiom AI Technologies is a fictional organization created for cybersecurity/GRC portfolio purposes. This document is a modeled demonstration and does not represent an actual organization's infrastructure, certification, audit opinion, production recovery capability, or confidential information.

1. Purpose

The purpose of this Disaster Recovery Plan (DRP) is to establish the technical framework for recovering critical information systems, infrastructure, applications, data, and supporting technology services following a disruptive event.

The DRP translates the recovery requirements identified in the Business Impact Analysis (BIA), BCP/DR Strategy, and Business Continuity Plan into a structured technical recovery approach.

The plan establishes disaster recovery activation criteria, technical recovery priorities, recovery roles, RTO/RPO requirements, decision and escalation thresholds, recovery sequencing, backup and restoration, validation, return to normal operations, testing, and continual improvement.

2. Scope

This plan applies to the technology services and infrastructure supporting Axiom AI Technologies' critical business activities.

Modeled scope includes cloud-hosted application infrastructure; AWS services; application services; PostgreSQL/RDS database infrastructure; identity and access; network and connectivity dependencies; backup and recovery services; security monitoring; logging and monitoring; third-party technology dependencies; and supporting development and deployment services.

It addresses infrastructure, database, cloud-service, application, data, backup, security-incident, access, and third-party technology disruptions. Detailed vendor-specific commands, credentials, infrastructure-as-code, and production secrets are intentionally excluded.

3. Relationship to Other Continuity Documents

The DRP is part of the broader Business Continuity and Disaster Recovery framework.

Business Impact Analysis → BCP/DR Strategy → Business Continuity Plan → Disaster Recovery Plan → Testing & Exercises → Lessons Learned & Continual Improvement.

4. Recovery Objectives

Recovery Time Objective (RTO): the target amount of time within which a system, service, or supporting capability should be restored following a disruption.

Recovery Point Objective (RPO): the target maximum amount of data loss, measured in time, that the organization can tolerate following a disruption.

Maximum Tolerable Period of Disruption (MTPD): the maximum period a business activity can remain unavailable before the resulting impact becomes unacceptable.

5. Technology Recovery Objectives

6. Disaster Recovery Roles

The roles below should be assigned to named individuals and backup personnel in an operational implementation.

7. DR Activation Criteria

The DRP may be activated when a critical production service is unavailable; a critical database is unavailable or corrupted; cloud infrastructure failure prevents normal service operation; a security incident requires restoration or rebuild; critical data cannot be accessed; backup restoration is required; a technology dependency threatens a P1 service; normal procedures cannot reasonably complete recovery; or an RTO/MTPD is threatened.

8. Disaster Recovery Levels

Recovery levels should be increased or decreased as additional information becomes available.

9. Decision and Escalation Thresholds

Decision and escalation thresholds provide objective criteria for determining when technical recovery requires additional resources, management involvement, or a change in recovery strategy.

Escalation should consider availability, expected recovery duration, RTO/RPO status, MTPD risk, data integrity, security implications, customer impact, dependencies, suppliers, personnel, and effectiveness of the current recovery approach.

When uncertainty exists, early escalation should be preferred where potential impact is significant.

10. RPO and Data-Recovery Thresholds

Data loss should be documented in terms of the applicable recovery point and business impact.

11. Technology Dependencies

The modeled Axiom AI environment contains several technology dependencies.

12. Recovery Sequence

The general recovery sequence may change depending on the nature of the incident.

13. Infrastructure Recovery

Infrastructure recovery should begin with assessment of affected cloud resources, availability zones or regions, network connectivity, compute, storage, security controls, configuration dependencies, and approved recovery configurations.

14. Identity and Access Recovery

Identity and access capabilities should be restored early because authorized personnel require secure access to recovery systems.

15. Database Recovery

The modeled primary database environment uses PostgreSQL/RDS.

Where data corruption is suspected, the recovery team should avoid overwriting potentially useful recovery sources until an appropriate recovery point has been identified and preserved.

16. Application Recovery

Application recovery should prioritize customer-facing and P1 functionality.

Non-critical application functionality may be deferred until P1 services have been stabilized.

17. Backup and Data Restoration

Backups are a critical component of disaster recovery. Backup-success notifications alone should not be treated as proof that restoration will succeed; restoration testing should be performed periodically.

18. Security Monitoring Recovery

Security visibility should be restored as early as practical during technical recovery.

19. Network and Connectivity Recovery

Network recovery should assess internet, application, database, administrative, DNS/service-discovery, firewall/security-group, and third-party connectivity.

20. Third-Party Recovery

Critical supplier dependencies should be reviewed as part of continuity and supplier-risk management.

21. Security Incident Recovery

When disaster recovery is triggered by a security incident, recovery must be coordinated with the Incident Response process.

Technical recovery should not inadvertently destroy evidence required for investigation.

Technology / Capability	Priority	RTO	RPO
SaaS Platform	P1	2 hours	4 hours
Security Monitoring	P1	2 hours	4 hours
PostgreSQL/RDS Database	P1	2 hours	4 hours
Identity & Access	P1	4 hours	24 hours
Customer Communications	P1	4 hours	24 hours
Development & Release Services	P2	8 hours	24 hours
Vulnerability Management Tooling	P2	8 hours	24 hours
Third-Party Security Management	P2	24 hours	48 hours
ISMS/GRC Systems	P3	72 hours	48 hours

These values represent portfolio assumptions and do not constitute verified technical capabilities.

Role	Responsibility
Executive Management	Strategic direction and major business-impact decisions.
Incident Lead	Coordinates incident and recovery.
Engineering / IT Lead	Coordinates technical recovery.
Security Lead	Security oversight and security-related recovery.
Database Owner	Database restoration and data-integrity validation.
Application Owner	Application recovery and functional validation.
Business Continuity / GRC Lead	Continuity requirements, evidence, and lessons learned.
Communications Lead	Internal and external recovery communications.
Process Owners	Validate business functionality after recovery.

Level	Description	Recovery Response
DR Level 1 — Local	Limited technical failure with minimal business impact.	Normal technical troubleshooting and restoration.
DR Level 2 — Significant	Important technology service affected.	Coordinated technical recovery with Incident Lead.
DR Level 3 — Major	P1 service or multiple critical systems affected.	Formal DR activation and coordinated recovery.

DR Level 4 — Severe	Major technology disruption requiring extensive recovery or alternate infrastructure.	Executive escalation and full DR coordination.
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Recovery Status	Threshold	Required Action
Within target	Recovery progressing within approved RTO.	Continue recovery and monitor.
At risk	Approximately 75% of RTO elapsed without sufficient progress.	Notify Incident Lead and technical owner; reassess.
Imminent breach	Approximately 90% of RTO elapsed without validated restoration.	Escalate and consider alternate recovery.
RTO exceeded	Approved RTO exceeded.	Formal escalation; document cause, impact, revised estimate.
MTPD at risk	Disruption could approach MTPD.	Executive escalation and continuity reassessment.

Condition	Required Response
Recovery point within RPO	Continue restoration and validate integrity.
Recovery point approaching RPO limit	Escalate to Database Owner and Engineering/IT Lead.
RPO exceeded	Document expected loss and escalate.
Required backup unavailable	Identify alternate recovery sources and escalate.
Backup integrity uncertain	Validate before relying on it.
Restored data integrity uncertain	Pause affected restoration where appropriate and investigate.

Dependency	Purpose	Recovery Consideration
AWS Cloud Infrastructure	Hosts critical services	Assess availability and regional/service dependency.
PostgreSQL/RDS	Database services	Validate backups, snapshots, and restoration.
Application Platform	SaaS functionality	Restore dependencies in appropriate sequence.
Identity & Access	Authentication/authorization	Required for authorized recovery.
Monitoring & Logging	Operational/security visibility	Prioritize sufficient visibility.
Backup Services	Data recovery	Validate availability and integrity.
Development/Deployment Services	Application recovery	Required for controlled restoration.
Third-Party Services	Supporting functionality	Escalate outages or use workarounds.

General recovery sequence:

1. Establish incident command.
2. Confirm personnel safety and access to required recovery resources.
3. Assess scope and impact.
4. Determine whether DR activation is required.
5. Protect evidence if a security incident is suspected.
6. Establish secure administrative access.
7. Restore foundational infrastructure and connectivity.
8. Restore identity and access capabilities.

9. Restore database services.
10. Restore application services.
11. Restore required monitoring and security visibility.
12. Validate data integrity.
13. Validate application functionality.
14. Validate security controls.
15. Confirm customer-facing availability.
16. Transition to normal operations.
17. Document recovery results and lessons learned.

Infrastructure Recovery Actions

18. Assess affected resources and dependencies.
19. Confirm approved recovery configurations.
20. Restore foundational infrastructure.
21. Validate network and security controls.
22. Restore required monitoring.
23. Validate availability before proceeding.

Identity and Access Recovery Actions

24. Confirm disruption and affected users.
25. Confirm authorized recovery personnel.
26. Establish approved emergency access if required.
27. Restore identity services.
28. Validate authentication and authorization.
29. Review and remove temporary access after recovery.

Database Recovery Actions

30. Identify failure scope.
31. Select the most appropriate validated recovery point.
32. Confirm recovery point against RPO.
33. Initiate approved restoration/failover.
34. Validate availability and integrity.
35. Confirm application connectivity.
36. Document the recovery point used.

Application Recovery Actions

37. Confirm dependencies.
38. Restore required infrastructure and database connectivity.
39. Deploy or restore the approved application version.
40. Validate configuration and authentication.
41. Perform functional and security checks.
42. Confirm customer-facing availability.

Backup and Data Restoration Actions

43. Identify required backup.
44. Validate availability/integrity where practical.
45. Restore to an appropriate recovery environment.
46. Validate completeness and integrity.
47. Obtain approval before production reconnection.
48. Monitor restored service.

Security Monitoring Recovery Actions

- 49. Identify lost visibility.
- 50. Restore logging and monitoring.
- 51. Validate alert generation and log ingestion.
- 52. Apply compensating measures if needed.
- 53. Document monitoring gaps.

Network and Connectivity Actions

- 54. Assess internet, application, database, administrative, DNS, firewall, and third-party connectivity.
- 55. Restore required connectivity.
- 56. Validate from administrative and application perspectives.

Third-Party Recovery Actions

- 57. Identify supplier impact.
- 58. Contact supplier through approved escalation path.
- 59. Obtain recovery estimate.
- 60. Compare estimate with RTO.
- 61. Identify workarounds or alternate provider.
- 62. Document supplier impact and decisions.

Security Incident Recovery Actions

- 63. Activate Incident Response.
- 64. Preserve evidence.
- 65. Contain affected systems as appropriate.
- 66. Identify known-good recovery point.
- 67. Rebuild/restore and apply security configuration.
- 68. Rotate/revoke compromised credentials where required.
- 69. Validate security and functionality before production return.

22. Actionable Recovery Runbooks

The following runbooks provide structured recovery procedures for common disaster scenarios. They guide authorized recovery personnel through triggers, assessment, actions, validation, escalation, and completion. Environment-specific commands and secrets are intentionally excluded.

22.1 — SaaS Application Outage

Recovery Objective: P1; RTO 2 hours; RPO 4 hours

- 70. Confirm outage and scope.
- 71. Check monitoring/service health.
- 72. Identify affected components and dependencies.
- 73. Record incident start time.
- 74. Notify Incident Lead and Application Owner.
- 75. Determine whether normal operations, application recovery, database recovery, or security response is required.
- 76. Restore known-good application version/configuration as appropriate.
- 77. Validate database connectivity, authentication, logging, monitoring, and critical customer workflows.
- 78. Escalate at 75%, 90%, RTO breach, data-integrity uncertainty, suspected compromise, or critical dependency failure.
- 79. Application Owner confirms functionality; Incident Lead authorizes return to normal.

22.2 — PostgreSQL/RDS Database Failure

Recovery Objective: P1; RTO 2 hours; RPO 4 hours

80. Confirm database availability and failure type.
81. Determine whether corruption is suspected.
82. Notify Database Owner and Incident Lead.
83. Select and preserve an appropriate validated recovery point.
84. Confirm it meets RPO where possible.
85. Restore/fail over using approved procedure.
86. Validate schema, data integrity, access controls, monitoring, and application connectivity.
87. Escalate if no suitable recovery point exists, RPO is threatened, integrity cannot be established, or RTO is threatened.

22.3 — Backup Restoration

Recovery Objective: Recovery point must meet applicable RPO

88. Identify data/system requiring restoration.
89. Determine required recovery point.
90. Identify and validate backup.
91. Authorize restoration.
92. Restore to recovery environment.
93. Validate data and integrity.
94. Obtain approval before production reconnection.
95. Monitor and document results.
96. Escalate if backup is unavailable/corrupt, RPO is exceeded, or restoration repeatedly fails.

22.4 — Identity and Access Recovery

Recovery Objective: P1; RTO 4 hours

97. Confirm identity-service disruption.
98. Identify affected administrative functions.
99. Confirm authorized recovery personnel.
100. Use approved emergency access if necessary.
101. Restore identity service.
102. Validate authentication, authorization, privileged access, and logging.
103. Remove temporary access after recovery.
104. Escalate for suspected compromise or unsafe administrative access.

22.5 — Security Incident Requiring System Recovery

Recovery Objective: Security/incident-response coordinated

105. Activate Incident Response and Security Lead.
106. Preserve evidence and determine isolation needs.
107. Identify known-good recovery point.
108. Restore/rebuild affected system.
109. Apply security configuration and required patches.
110. Rotate/revoke compromised credentials where required.
111. Restore dependencies and monitoring.
112. Perform security and functional validation.
113. Return to production only after authorization.

22.6 — AWS Infrastructure Failure

Recovery Objective: Assess service/resource/zone/region impact

114. Identify affected AWS services.
115. Assess scope and customer impact.
116. Confirm administrative access.
117. Determine whether recovery is possible in current environment.
118. Use approved alternate infrastructure/recovery capability if required.
119. Restore network, compute, database connectivity, security controls, and monitoring.
120. Validate application functionality.
121. Escalate if P1 RTO is threatened or alternate-region/major decisions are required.

22.7 — Monitoring and Logging Recovery

Recovery Objective: Restore sufficient operational/security visibility

122. Identify lost visibility.
123. Identify alternate telemetry.
124. Restore logging and monitoring.
125. Validate alerting and ingestion.
126. Increase manual monitoring or restrict unnecessary admin activity if required.
127. Document visibility gaps and confirm Security Lead acceptance.

22.8 — Third-Party Technology Outage

Recovery Objective: Compare supplier estimate to affected RTO

128. Identify supplier and affected services.
129. Contact supplier through approved path.
130. Obtain recovery estimate.
131. Compare estimate with RTO.
132. Identify workarounds or alternate provider.
133. Monitor supplier communications.
134. Escalate for P1 impact, no workaround, poor communication, or contractual/security concerns.

22.9 — Recovery Validation and Service Return

Recovery Objective: Complete after technical recovery

135. Confirm infrastructure and database availability.
136. Confirm application, authentication, authorization, and critical workflows.
137. Validate data integrity, security controls, logging, monitoring, backups, and dependencies.
138. Obtain Process Owner validation.
139. Record actual recovery time and recovery point.
140. Communicate restoration and transition to normal operations.

23. Recovery Validation

A system is not fully recovered merely because it is technically online.

Validation Area	Acceptance Check
Availability	Accessible to authorized users.
Functionality	Critical business functions operate as expected.
Data Integrity	Recovered information is complete and consistent with selected recovery point.

Security	Required security controls are operating.
Monitoring	Logging and monitoring are functioning.
Connectivity	Required dependencies communicate successfully.
Business Validation	Process Owner confirms service supports the required business activity.

24. Recovery Acceptance Criteria

- Critical infrastructure is operational.
- Required databases are available.
- Application functionality has been validated.
- Authentication and authorization are functioning.
- Data integrity has been confirmed.
- Required security monitoring is operational.
- Critical dependencies are available or mitigated.
- Process Owners confirm business functionality.
- Temporary emergency measures are documented.
- Outstanding risks are understood and accepted by the appropriate authority.

25. Return to Normal Operations

141. Confirm stable operation.
142. Confirm security monitoring.
143. Remove unnecessary emergency access.
144. Reconcile temporary configurations.
145. Confirm normal backup operations.
146. Confirm normal monitoring and alerting.
147. Close temporary workarounds.
148. Confirm business acceptance.
149. Communicate restoration.
150. Document outstanding issues.

26. Post-Recovery Review

- Cause and scope.
- Systems and business activities affected.
- Actual recovery time and recovery point.
- RTO/RPO performance.
- Recovery sequence and dependencies.
- Backup performance.
- Security controls.
- Communications and supplier performance.
- Personnel availability.
- Decision-making and escalation.
- Corrective actions.

27. Testing and Exercises

- Tabletop exercises validate roles, escalation, decisions, communications, and sequencing.
- Backup restoration tests validate backup availability, restoration, integrity, and recovery-point assumptions.
- Technical recovery tests validate infrastructure, application, database recovery, and RTO assumptions.
- Failover/recovery exercises, where appropriate, validate alternate recovery capabilities.
- Testing results should be documented and tracked through corrective actions.

28. Recovery Evidence

- Incident timeline.
- Recovery decision log.
- Recovery checklist.
- Backup restoration results.
- System recovery records.
- Validation results.
- RTO/RPO measurements.
- Communications.
- Technical logs.
- Recovery validation evidence.
- Post-recovery review.
- Corrective-action records.

29. DRP Maintenance

- Review at least annually.
- Review after significant recovery events or tests.
- Review after major infrastructure/application architecture changes.
- Review after significant supplier changes.
- Review after material RTO/RPO changes.
- Review after significant security incidents.
- Version-control approved changes.

30. Document Ownership and Review

Role	Responsibility
Document Owner — Information Security / GRC	Maintains DRP, coordinates reviews, tracks revisions, and addresses documentation gaps.
Technical Owner — Engineering / IT	Validates technical recovery approach against actual architecture in an operational implementation.
Business Owner — Executive Management	Provides oversight and approves significant continuity/recovery changes.
Process Owners	Validate business and technical recovery requirements.
Approval Authority — Executive Management	Approves the plan and significant changes.

Review should confirm technology dependencies, recovery priorities, RTO/RPO, roles, technical procedures, backup assumptions, security requirements, suppliers, and corrective actions.

31. Disaster Recovery Activation Checklist

Step	Action	Status
1	Identify technology disruption	☐
2	Assess business and technical impact	☐
3	Determine DR activation level	☐
4	Establish Incident Lead and recovery team	☐
5	Preserve evidence where required	☐
6	Confirm secure administrative access	☐
7	Identify required recovery point	☐
8	Restore foundational infrastructure	☐
9	Restore identity and access	☐
10	Restore database services	☐
11	Restore application services	☐
12	Restore monitoring and security visibility	☐
13	Validate data integrity	☐
14	Validate application functionality	☐
15	Validate security controls	☐
16	Obtain business validation	☐
17	Communicate restoration	☐
18	Remove temporary recovery measures	☐
19	Transition to normal operations	☐
20	Conduct post-recovery review	☐
21	Track corrective actions	☐

32. Related Documents

- Business Impact Analysis.
- BCP/DR Strategy.
- Business Continuity Plan.
- Risk Register.
- Statement of Applicability.
- Information Security Policy.
- Access Control Policy.
- Incident Response Policy.
- Asset Management Policy.
- Third-Party Security Policy.
- Tabletop Exercise Materials.
- Recovery Test Results.

33. Portfolio Disclaimer

This Disaster Recovery Plan is a fictional cybersecurity/GRC portfolio artifact created to demonstrate practical application of disaster recovery planning principles. The infrastructure, technologies, recovery objectives,

procedures, dependencies, thresholds, and assumptions are modeled examples. They do not represent verified production capabilities of a real organization.

A real-world Disaster Recovery Plan would require validation against the organization's actual architecture, contracts, technical configurations, backup capabilities, recovery tooling, personnel, and business requirements.

34. Document Approval

Role	Name	Approval Date	Status
Document Owner — Information Security / GRC	TBD	TBD	Pending
Technical Owner — Engineering / IT	TBD	TBD	Pending
Business Owner — Executive Management	TBD	TBD	Pending

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